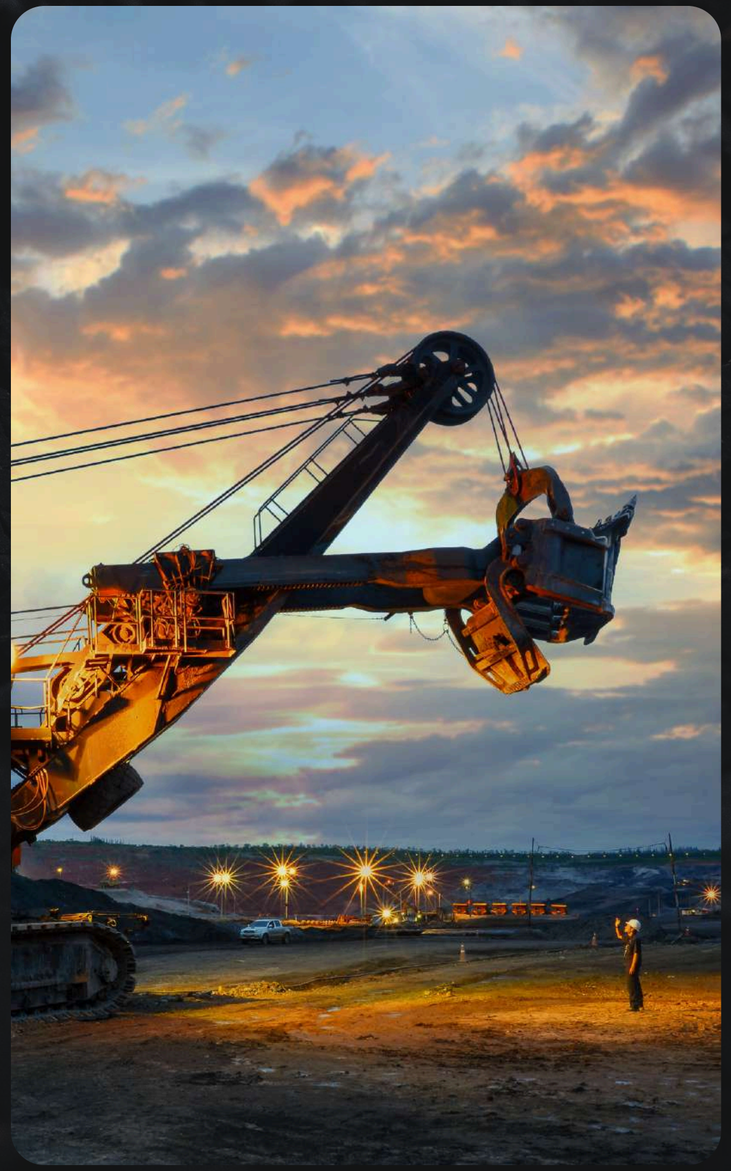

GOAL OF THE STUDY

The goal of this study is to find which macroeconomic variables impact more the mining industry of Chile, focusing on copper: prices, gdp, resource paradox, labour, exports, imports and global shocks.

INDEX

1. Socio-Economics Dimensions and Labour Dynamics

2. The Copper Dependence: Economic & Social Impact on Chile

3. International Market and Geopolitical Drivers of the Chilean Copper Industry

4. LME and COMEX indexes: analysis and forecasting

1. Socio-Economics Dimensions and Labour Dynamics

A statistical investigation into labor trends, income inequality, and human development in Chile's mines (2005-2024)

- Labor & Price
- Wealth distribution
- Social impact

The "COCHILCO's Anuario de Estadísticas del Cobre y Otros Minerales " dataset

TABLA / TABLE 31.1

PERSONAL OCUPADO EN MINERÍA[1]

Employment in the Mining Sector[1]

(Promedio Anual de Trabajadores) / (Yearly Average)

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
A. TRABAJADORES EN FAENAS MINERAS POR GÉNERO / EMPLOYEES IN MINING OPERATIONS BY GENDER										
MUJERES / WOMEN	**5.893**	**6.322**	**7.953**	**9.409**	**10.438**	**11.760**	**12.557**	**17.205**	**17.061**	**18.918**
Participación en el total / total share	4,40%	4,76%	5,19%	5,71%	5,99%	6,16%	6,37%	7,27%	7,44%	7,57%
HOMBRES / MEN	**128.098**	**126.485**	**145.399**	**155.233**	**163.687**	**179.283**	**184.640**	**219.566**	**212.314**	**230.897**
Participación en el total / total share	95,60%	95,24%	94,81%	94,29%	94,01%	93,84%	93,63%	92,73%	92,56%	92,43%
TOTAL PERSONAL EN MINERÍA / Total Employment in Mining	**133.991**	**132.807**	**153.352**	**164.642**	**174.125**	**191.043**	**197.197**	**236.771**	**229.375**	**249.815**
B. TRABAJADORES EN FAENAS MINERAS SEGÚN TIPO DE EMPRESA / EMPLOYEES IN MINING OPERATIONS BY COMPANY										
MANDANTE / OWNER	**48.098**	**47.913**	**54.434**	**58.470**	**57.901**	**58.699**	**61.849**	**70.037**	**74.214**	**71.080**
Participación en el total / total share	35,90%	36,08%	35,50%	35,51%	33,25%	30,73%	31,36%	29,58%	32,35%	28,45%
CONTRATISTA / CONTRACTOR	**85.891**	**86.052**	**98.918**	**106.173**	**116.225**	**132.345**	**135.349**	**166.737**	**155.114**	**178.735**
Participación en el total / total share	64,10%	64,79%	64,50%	64,49%	66,75%	69,27%	68,64%	70,42%	67,62%	71,55%
Ajuste[2] / Adjustment	**2**	**-1.158**	**-**	**-0**	**-0**	**-1**	**-0**	**-3**	**47**	**-**
Participación en el total / total share	**0,00%**	**-0,87%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,02%**	**0,00%**
TOTAL PERSONAL EN MINERÍA / Total Employment in Mining	**133.991**	**132.807**	**153.352**	**164.642**	**174.125**	**191.043**	**197.197**	**236.771**	**229.375**	**249.815**
C. TOTAL OCUPACIÓN PAÍS / Total Country Employment	**6.273.620**	**6.410.980**	**6.567.240**	**6.740.410**	**6.901.920**	**7.572.318**	**7.786.902**	**7.952.257**	**8.149.389**	**8.153.126**
D. MINERÍA EN EL PAÍS / Mining, Overall (%)	**2,14%**	**2,07%**	**2,34%**	**2,44%**	**2,52%**	**2,60%**	**2,53%**	**2,98%**	**2,81%**	**3,06%**

COCHILCO
Ministerio de Minería

Gobierno de Chile

COCHICLO raw data: mining employment 2005-2024

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Participación en el total / total share	35,90%	36,08%	35,50%	35,51%	33,25%	30,73%	31,36%	29,58%	32,35%	28,45%
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Participación en el total / total share	64,10%	64,79%	64,50%	64,49%	66,75%	69,27%	68,64%	70,42%	67,62%	71,55%
Ajuste[2] / Adjustment	**2**	**-1.158**	**-**	**-0**	**-0**	**-1**	**-0**	**-3**	**47**	**-**
Participación en el total / total share	**0,00%**	**-0,87%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,00%**	**0,02%**	**0,00%**
TOTAL PERSONAL EN MINERÍA / Total Employment in Mining	**133.991**	**132.807**	**153.352**	**164.642**	**174.125**	**191.043**	**197.197**	**236.771**	**229.375**	**249.815**
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COCHILCO Annual Statistics (Tables 31.1)

TABLA / TABLE 31.2

PERSONAL OCUPADO EN MINERÍA[1]

Employment in the Mining Sector[1]

(Promedio Anual de Trabajadores) / (Yearly Average)

	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
A. TRABAJADORES EN FAENAS MINERAS POR GÉNERO / EMPLOYEES IN MINING OPERATIONS BY GENDER										
MUJERES / WOMEN	**18.839**	**17.482**	**17.615**	**19.284**	**21.455**	**22.045**	**27.715**	**32.683**	**36.292**	**38.590**
Participación en el total / total share	7,90%	8,01%	8,10%	8,45%	8,62%	9,38%	10,06%	10,50%	11,02%	11,77%
HOMBRES / MEN	**219.615**	**200.678**	**199.789**	**209.056**	**227.348**	**212.919**	**247.860**	**278.608**	**293.001**	**289.195**
Participación en el total / total share	92,10%	91,99%	91,90%	91,55%	91,38%	90,62%	89,94%	89,50%	88,98%	88,23%
TOTAL PERSONAL EN MINERÍA / Total Employment in Mining	**238.454**	**218.160**	**217.404**	**228.340**	**248.803**	**234.964**	**275.575**	**311.291**	**329.293**	**327.785**
B. TRABAJADORES EN FAENAS MINERAS SEGÚN TIPO DE EMPRESA / EMPLOYEES IN MINING OPERATIONS BY COMPANY										
MANDANTE / OWNER	**70.877**	**66.250**	**65.392**	**67.375**	**67.187**	**65.812**	**67.534**	**68.717**	**71.636**	**71.272**
Participación en el total / total share	29,72%	30,37%	30,08%	29,51%	27,00%	28,01%	24,51%	22,07%	21,75%	21,74%
CONTRATISTA / CONTRACTOR	**167.577**	**151.910**	**152.012**	**160.965**	**181.616**	**169.152**	**208.041**	**242.574**	**257.657**	**256.513**
Participación en el total / total share	70,28%	69,63%	69,92%	70,49%	73,00%	71,99%	75,49%	77,93%	78,25%	78,26%
TOTAL PERSONAL EN MINERÍA / Total Employment in Mining	**238.454**	**218.160**	**217.404**	**228.340**	**248.803**	**234.964**	**275.575**	**311.291**	**329.293**	**327.785**
C. TOTAL OCUPACIÓN PAÍS / Total Country Employment	**8.293.054**	**8.389.930**	**8.592.194**	**8.783.603**	**8.975.910**	**7.922.464**	**8.313.958**	**8.862.696**	**9.074.389**	**9.298.541**
D. MINERÍA EN EL PAÍS / Mining, Overall (%)	**2,88%**	**2,60%**	**2,53%**	**2,60%**	**2,77%**	**2,97%**	**3,31%**	**3,51%**	**3,63%**	**3,53%**

COCHILCO Annual Statistics (Tables 31.2)

Number of Chile mining workforce

Current Period (2021-2024) is the highest employment era in Chilean history.
Structural Change: Since 2021, the workforce has detached from historical levels, growing 148.2% since 2005.

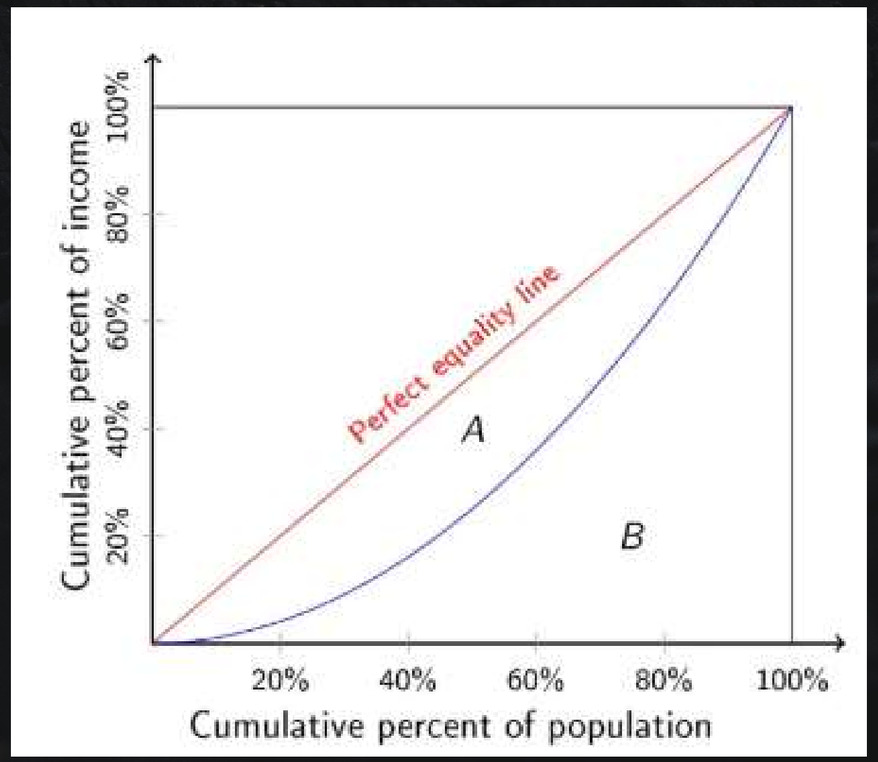
```python
# --- PLOT 1: THE 20-YEAR EVOLUTION ---
ax1 = fig.add_subplot(gs[0, :])
ax1.fill_between(df['Year'], df['Workforce'], color='teal', alpha=0.1)
ax1.plot(df['Year'], df['Workforce'], color='teal', marker='o', linewidth=2, label='Total Workforce')
ax1.plot(df['Year'], df['Workforce_MA3'], color='orange', linestyle='--', label='3-Year Trend')

# Highlight Key Historical Events
events = {
    2009: "Global Financial Crisis",
    2012: "First Historical Peak",
    2016: "Price Slump",
    2020: "COVID-19 Dip",
    2023: "EV / Green Boom"
}
for yr, txt in events.items():
    ax1.annotate(txt, xy=(yr, df.loc[df['Year']==yr, 'Workforce'].values[0]),
                 xytext=(0, 20), textcoords='offset points', arrowprops=dict(arrowstyle='->'), ha='center')

ax1.set_title('Two Decades of Chile Mining Workforce (2005-2024)', fontsize=16, fontweight='bold')
ax1.set_ylabel('Number of Employees')
ax1.legend()
```

The Gini coefficient

The Gini Coefficient is the international standard indicator for measuring inequality in income distribution.

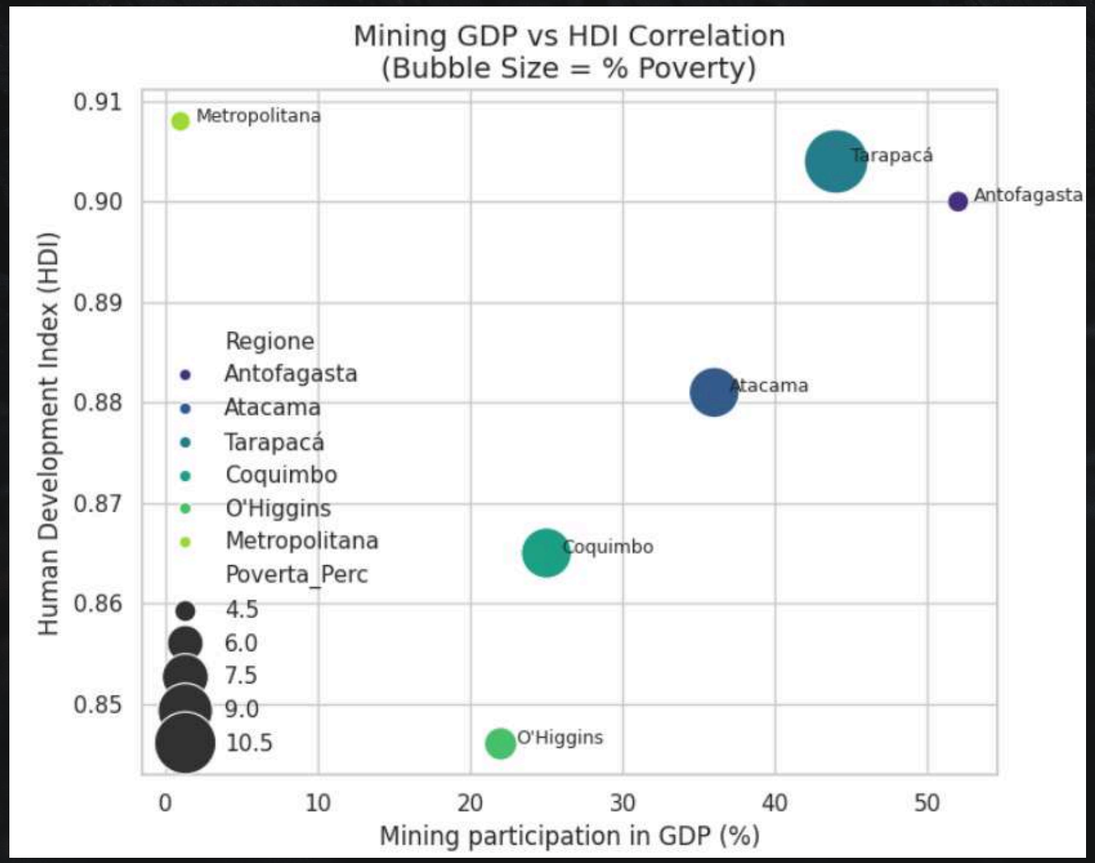


The index ranges from 0: perfect equality (everyone has the same income) to 1 : perfect Inequality (a single individual earns the total income of the economy)

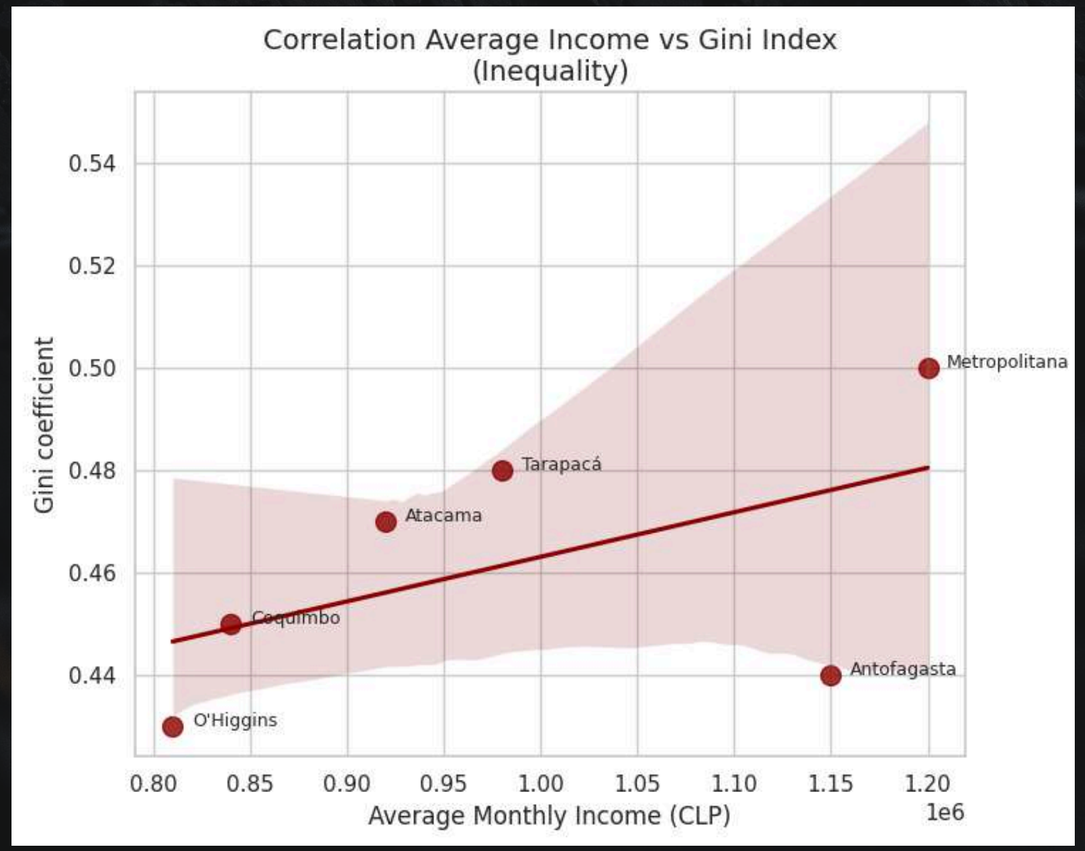
High GDP per capita in mining regions (like Antofagasta) often masks deep social gaps. The Gini Index allows us to "penalize" nominal wealth to find the real welfare.

Mining-heavy regions show Gini coefficients between 0.44 and 0.48.
These values indicate a "High Inequality" trap: even when the economy grows due to high copper prices, the distribution remains rigid, preventing the HDI from reaching its full potential.

Mining GDP vs HDI correlation



Correlation Average Income vs Gini Index



The "Resource Paradox"

Key macroeconomic insights

Structural Wage Divergence: Real mining wages grew > 5 ince 2005, creating a a tany a high-income labor elite bu but wideiing the inter-sectoral gap

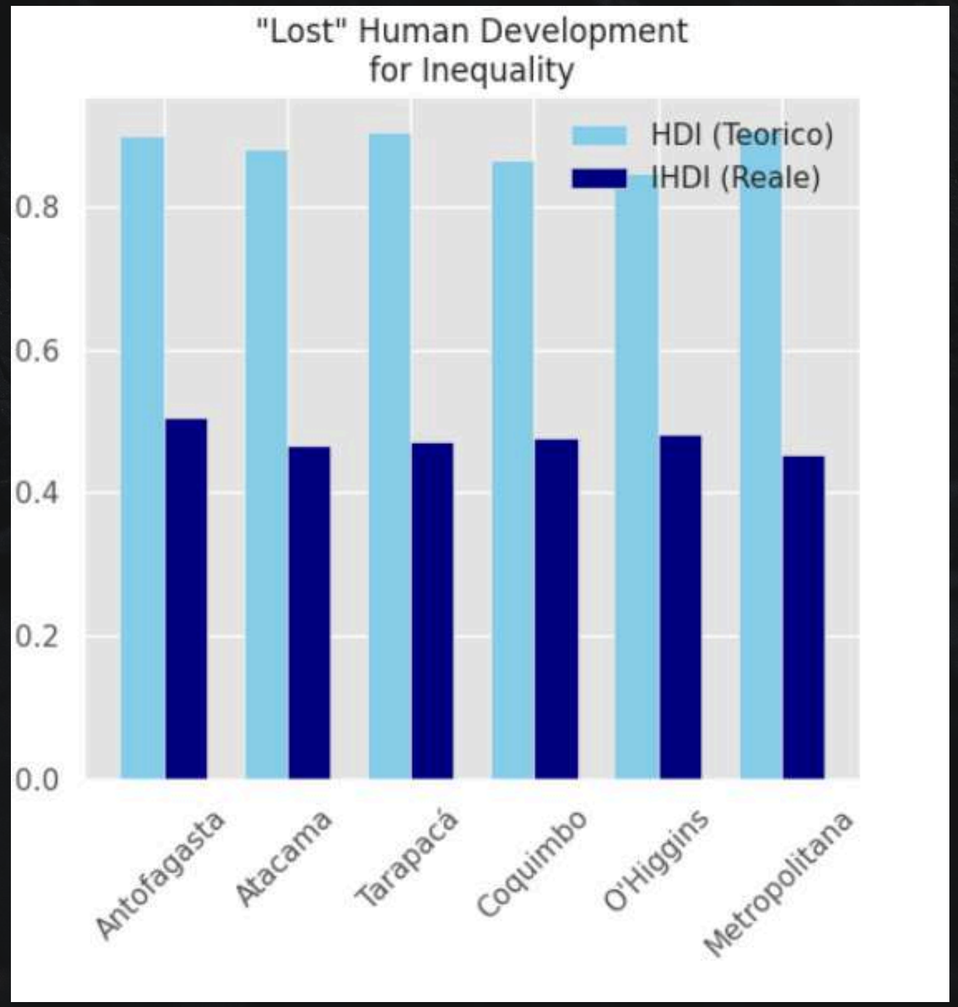
The Inequality Trap (Gini vs. Income): Higher GDP in mining regions with high Gini indices (0.44-50), concentating wealth in capital intanstive operations

Operational Resillience is driven by "structural necesiity" (decliining ore grades, deeper mines) trather short-thorm price volablity

Regional performance matrix

Indicator	Mining-Heavy Regions (North)	National Average / Metropottan
HDI	Higher	Stable
Poverty %	Persistent (pockets)	Moderate
Gini Index	High (0.44-550)	High (Structural)

"Lost" Human Development for Inequality



$$IHDI = HDI \times (1 - Gini)$$

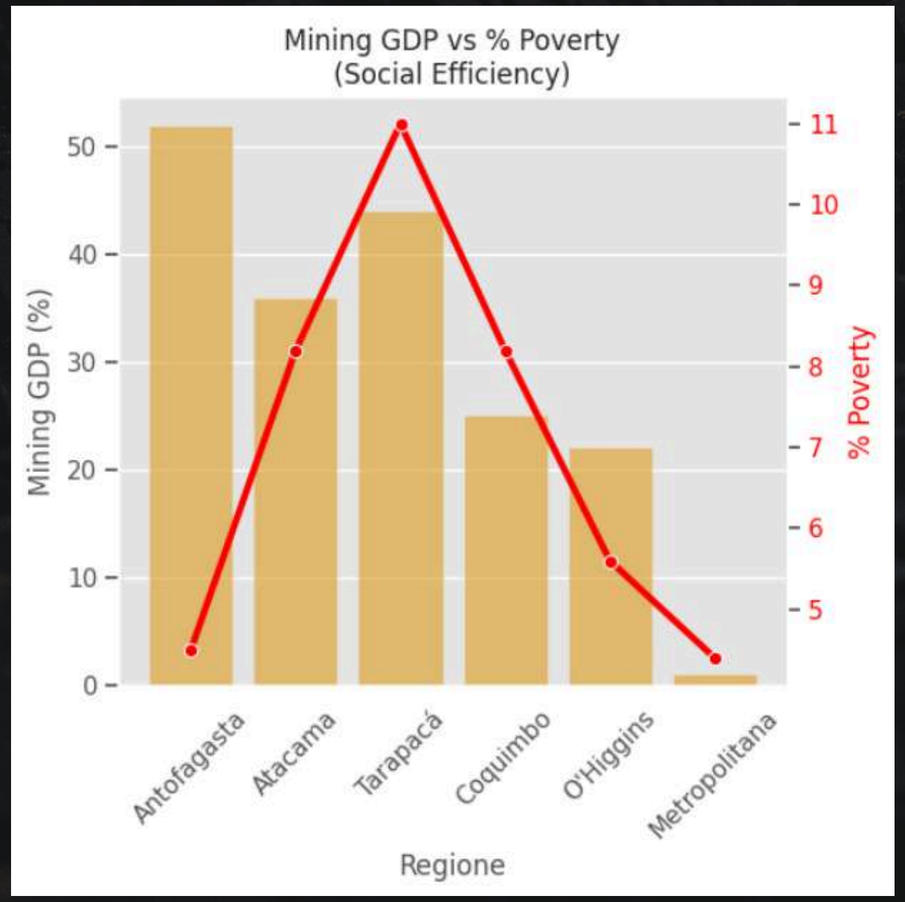
The difference between HDI and IHDI represents the "Lost Development" due to unequal resource distribution.

```
plt.subplot(1, 3, 1)
x = np.arange(len(df['Regione']))
width = 0.35

plt.bar(x - width/2, df['HDI'], width, label='HDI (Teorico)', color='skyblue')
plt.bar(x + width/2, df['IHDI'], width, label='IHDI (Reale)', color='navy')

plt.title('"Lost" Human Development\nfor Inequality', fontsize=12)
plt.xticks(x, df['Regione'], rotation=45)
plt.legend()
```

Mining GDP vs % Poverty (Social Efficency)




```
ax2 = plt.subplot(1, 3, 2)
ax2_line = ax2.twinx()

sns.barplot(x='Regione', y='PIL_Minerario_Perc', data=df, ax=ax2, color='orange', alpha=0.6)
sns.lineplot(x='Regione', y='Poverta_Perc', data=df, ax=ax2_line, color='red', marker='o', linewidth=3)

ax2.set_title('Mining GDP vs % Poverty\n(Social Efficiency)', fontsize=12)
ax2.tick_params(axis='x', rotation=45)
ax2.set_ylabel('Mining GDP (%)')
ax2_line.set_ylabel('% Poverty', color='red')
ax2_line.tick_params(axis='y', labelcolor='red')
ax2_line.grid(False)
```

This metric evaluates the ability of a region to convert industrial wealth (Mining GDP) into poverty reduction. High "Social Efficiency" occurs when mining profits are successfully redistributed through local infrastructure and social programs. We observe a "Poverty Paradox" in high-output regions like Tarapaca. Despite contributing significantly to the national GDP, poverty levels remain persistent due to the Enclave Effect, where the mining sector operates as an isolated wealth generator with limited spillover into the local non-mining economy.